Aymen Akdim



Embedded Electronics Master's — PFE Candidate, Embedded & Automotive

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PROFILE

Master's student in Embedded Electronics (Ibn Tofail University, Faculty of Sciences) with a Bachelor's degree in Electrical Engineering, Instrumentation & Supervision. Passionate about automotive embedded software, I develop in C/C++ on microcontrollers and Raspberry Pi, and I built an ECU network simulator communicating over a CAN bus with OBD-II diagnostics. Used to testing, debugging and documenting my own developments, I am looking for an **6-month end-of-studies internship (PFE) starting in January 2027** in embedded systems to contribute to the development, integration and validation of systems for mobility.

KEY SKILLS

Embedded programming: C (C11), C++, Python, MATLAB / Simulink, VHDL / Verilog
Microcontrollers & architectures: Arduino, ESP32, 8051, Raspberry Pi 4, FPGA
Protocols & automotive: CAN (8-byte frames, encoding/decoding), ECU, OBD-II diagnostics (DTC), ADAS, I2C, MQTT
Testing, validation & tools: pytest (automated testing), Git / GitHub, Proteus, TIA Portal & HMI, Grafana, Node-RED
Other: Image processing, Flutter, Firebase RTDB, MS Office

PROJECTS

Automotive ECU Network & Diagnostic Simulator — *Personal project*

C11 · GCC (MSYS2) · Git/GitHub · Python/pytest

- Developed in C a simulator of an automotive ECU network (engine, dashboard) communicating over a CAN bus.
- Virtual engine ECU: sensor validation, 10 Hz control cycle, diagnostic trouble code (DTC OBD-II) management.
- CAN frame encoding/decoding (8 bytes, big-endian, scaling factors) within a modular, testable architecture.
- Preparing fault injection and automated functional tests with Python/pytest.

Traffic Sign Detection (ADAS) — *Personal project*

Raspberry Pi 4 Model B · Image processing

- Embedded system for real-time road sign detection and recognition (camera capture, preprocessing, classification).
- Validated in real-world conditions (lighting, distance, viewing angle) and analyzed failure cases.

Smart Soap Dispenser (IoT) — *Personal project*

Microcontroller · Firebase Realtime Database · Flutter · Web

- Full integration: sensors/actuators → microcontroller → cloud → web and mobile application.
- Real-time supervision (distribution data, equipment status) through a Flutter app and a website synchronized with Firebase.

FPGA Simulation and Programming — *Academic project*

VHDL · Verilog

- Simulation of digital circuits on an FPGA board; design and simulation of a simplified RISC microprocessor.

EXPERIENCE

Electrical Maintenance Intern — OCP — *1 month · 2024*

- Diagnosis and corrective maintenance of electric motors (disassembly, replacement of bearings, windings and brushes).
- Low- and high-voltage electrical tests (insulation, dielectric strength, load tests): measurement rigor and failure analysis.

EDUCATION

Master's in Embedded Electronics — *2025 – 2027*

Ibn Tofail University, Faculty of Sciences, Kénitra · ongoing

Bachelor's in Electrical Engineering, Instrumentation and Supervision of Smart Systems — *2024 – 2025*

EST, Kénitra

Diploma in Electromechanics of Automated Systems — *2022 – 2024*

ISTA, Khouribga

LANGUAGES & INTERESTS

Arabic (native) · **French** (fluent · B1) · **English** (fluent · B2)
Interests: Strategy video games, Volleyball